

CONTACT

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EDUCATION

MASTER OF SCIENCE IN AI AND ML

2026-2027

- LiverPool John Moores University, London
- Pursuing

POST GRADUATE IN AI AND ML

2023-2025

- International Institute of Information Technology, Bangalore
- Passed with first class distinction (95%)

BACHELOR OF ENGINEERING, ELECTRONICS AND INSTRUMENTATION

2016-2020

- St.Jospehs College of Engineering
- Passed with first class distinction (86%)

SKILLS

- Programming Languages: Python (Expert), C++
- Deep Learning Frameworks: TensorFlow, PyTorch, Keras
- Computer Vision Libraries: OpenCV, Scikit-image, Pillow

KAVIYA SUBRAMANIAN

AI/ML ENGINEER

PROFILE

Results-driven AI Engineer with 5 years of expertise designing, deploying, and leading robust CV and Generative AI solutions. Proven track record in deep learning (object detection, tracking), alongside spearheading structured RAG architectures and LLM orchestration to optimize operational efficiency. Currently pursuing MSc in AI/ML in LJMU. Located in the Netherlands with unrestricted work authorization (no visa required), seeking impactful AI/ML Engineering roles.

WORK EXPERIENCE

Master works 2024 - PRESENT

Computer Vision Engineer

- Led a 2-member engineering team through the end-to-end development, delivery, and continuous maintenance of production-grade computer vision systems and tracking algorithms
- Developed and deployed computer vision models for vehicle inspection in the SASO project.
- Developed object detection and License Plate Recognition(LPR) model with more than 95% accuracy
- Optimized performance using DeepStream and integrated visual language models (VLM) that increased accuracy by 10%.
- Collaborated with cross-functional teams for seamless integration of solutions.
- Co-developed a Text-to-SQL RAG agent using LangChain and PostgreSQL, enabling natural language querying of the SASO vehicle database
- Implemented Few-Shot Prompting within the LangChain agent to provide the LLM with schema contexts, successfully mitigating query hallucinations and ensuring accurate data retrieval
- Conducted root-cause analysis on failed queries and optimized the mapping of complex database schemas to improve the reliability of human-readable responses

E-con systems (Visai Labs) 2020 - 2024

Computer Vision Engineer

- Developed real-world solutions including V-MEASURE for logistics, featuring an image classification model with 99% accuracy, reducing measurement errors by 30%.
- Created proposals for deep learning improvements in existing products.
- In RETAILYTIX, implemented person tracking with SOTA algorithms and a visualization tool for CCTV, along with a YOLO head detection model achieving 98% accuracy to assess retail inflow.
- Built a Python backend for a data analytics dashboard, boosting customer traction by over 20%.

- **Machine Learning Libraries:** Scikit-learn, Pandas, NumPy, Matplotlib
- **Cloud Platforms:** AWS (S3, EC2, SageMaker)
- **MLOps & Deployment:** Docker, Kubernetes (Basic/Familiarity), Flask, Streamlit, Git, CI/CD Concepts
- **Databases & Querying:** MySQL
- **Core AI/ML Concepts:** Deep Learning, Machine Learning, Supervised Learning, Object Detection, Image Classification, Semantic Segmentation, Pose Estimation, Defect Detection, Data Augmentation
- **Generative AI:** LLM, LangChain, LlamaIndex, Agentic AI, VLMs
- **RAG Architectures:** Text-to-SQL, Vector Search, Few-Shot Prompting

KEY ACHIEVEMENTS

- Received Employee of the month for November 2025.
- Received the 2022 Budding Engineer Award for impactful projects.
- Two-time recipient of the Extra Mile Award for hard work in quarterly meetings.

LANGUAGES

- English (Fluent)
- Tamil (Fluent)
- Hindi (Basics)
- Dutch (Basics)

- Collected and annotated project data using SOTA tools, cutting operation time by 30%, and utilized DVC for data versioning.
- Developed internal analysis tools in Python, enhancing metric evaluation and increasing productivity by 25%.
- For V-TRACK, improved head tracking accuracy by 20% through an image classification model and streamlined result analysis, reducing delivery time by over 50%.
- In VIUDIENCE, worked on a retail project employing models for face detection and emotion prediction, contributing to business improvement analysis.
- Created business logic for determining person height and emotion dwell time, and developed a Qt desktop application for product demonstration, increasing traction.

PROJECTS

- **YOLOv8 Object Detection App (Image & Real-time)**
 - Developed a YOLOv8 detection inference engine using Flask to enable high-performance object detection for both static images and real-time video streams.
 - [Github link Image](#) [Github link Real time](#)
- **Vehicle Inspection System** Masterworks
 - Engineered an automated inspection pipeline using YOLO, DeepStream, and License Plate Decoding, integrated into CI/CD workflows for reliable production deployments.
- **Retail Analytics (Retailytix)** VisAi Labs
 - Implemented multi-camera person tracking using Custom YOLO and DeepSort to facilitate customer behavior analysis and real-world spatial mapping.
- **V-TRACK: Traffic Monitoring** VisAi Labs
 - Developed a single-camera head tracking system for closed environments like lifts to monitor traffic flow using OpenCV and TensorFlow.
- **Predictive ML & Financial Analytics**
 - Built robust predictive models for financial defaulter analysis and housing price forecasting using Scikit-learn, Ridge/Lasso Regression, and RFE.
 - [Lending Club Case Study](#)
 - [Bike Sharing Company Analysis](#)
 - [Housing Price Prediction](#)